

Sales Analysis Dashboard Using Tableau

Project Overview

This project focuses on analyzing **sales, customer, payment, category, gender, invoice, and shopping mall data using Tableau**. An interactive dashboard was created to convert raw sales data into meaningful business insights.

The dashboard helps understand **overall sales performance, product category contribution, customer demographics, payment methods, monthly sales trends, and mall-wise sales performance**.

Objectives

- Analyze overall sales performance.
- Track important sales KPIs.
- Compare sales across product categories.
- Analyze sales by gender.
- Analyze sales by payment method.
- Identify top-performing shopping malls.
- Study monthly sales trends.
- Analyze customer purchasing behavior.
- Build an interactive Tableau dashboard.
- Present business insights through data visualization.

Dataset Information

The dashboard uses customer shopping and sales information containing fields such as:

- Invoice Number
- Invoice Date
- Customer ID
- Gender
- Age
- Category
- Quantity
- Price
- Payment Method
- Shopping Mall
- Sales

Tools & Technologies

- **Tableau Public**
- Data Visualization
- Calculated Fields
- Filters
- Parameters
- Dashboard Design
- Interactive Charts
- KPI Cards
- Excel/CSV Dataset

KPIs Created

The dashboard contains the following key performance indicators:

KPI	Value
Total Sales	251,505,794
Total Quantity	298,712
Average Order Value (AOV)	2,529
Total Invoices	99,457

AOV Formula

`AOV = Total Sales / Total Invoices`

AOV represents the average amount spent per invoice/order.

Dashboard Visualizations

1. Sales by Category

A **Treemap** is used to compare sales contribution across different product categories.

Categories visible in the dashboard include:

- Clothing
- Shoes
- Technology
- Other categories

This visualization makes it easy to identify which categories contribute the most to total sales.

2. Sales by Gender

A **Pie Chart** is used to compare sales between male and female customers.

The dashboard shows approximately:

- Female: **59.72%**
- Male: **40.28%**

This helps understand the gender distribution of sales.

3. Sales by Payment Mode

A **Pie Chart** displays sales distribution by payment method.

Payment methods include:

- Cash
- Credit Card
- Debit Card

The dashboard shows:

- Cash: **44.86%**
- Credit Card: **35.02%**
- Debit Card: **20.12%**

4. Sales by Invoice Month

A **Line Chart** is used to analyze sales trends over invoice dates/months.

This helps identify:

- High-sales periods
- Low-sales periods
- Sales fluctuations
- Overall sales trends

5. Sales by Shopping Mall

A **Horizontal Bar Chart** compares sales across shopping malls.

The dashboard highlights malls such as:

- Mall of Istanbul
- Kanyon
- Metrocity
- Metropol AVM
- Istinye Park
- Zorlu Center
- Cevahir AVM
- Viaport Outlet
- Emaar Square Mall
- Forum Istanbul

This helps identify the best-performing shopping locations.

Analysis Performed

Sales Analysis

- Total sales calculation
- Sales by category
- Sales by shopping mall
- Monthly sales trend
- Sales distribution by gender
- Sales distribution by payment method

Customer Analysis

- Gender-wise sales contribution
- Customer purchasing behavior
- Invoice/customer transaction analysis

Payment Analysis

- Cash transaction analysis
- Credit card transaction analysis
- Debit card transaction analysis

Category Analysis

- Category-wise sales contribution
- Identification of high-performing categories
- Comparison of product categories

Mall Analysis

- Mall-wise sales comparison
- Identification of top-performing malls
- Comparison of sales performance across locations

Tableau Concepts Used

- Data Connection
- Data Preparation
- Dimensions & Measures
- Calculated Fields
- KPI Cards
- Treemap
- Pie Chart
- Line Chart
- Bar Chart
- Filters
- Dashboard
- Formatting
- Sorting
- Labels
- Tooltips
- Interactive Visualization
- Dashboard Layout

Key Business Insights

Based on the dashboard:

- **Clothing** is one of the strongest sales-contributing categories.
- **Female customers contribute a higher share of sales** than male customers.
- **Cash** is the largest payment method by sales contribution.
- **Mall of Istanbul and Kanyon** are among the strongest-performing shopping malls.
- Sales show noticeable variation across invoice months.
- Category and mall analysis can help management identify high-performing business areas.
- The dashboard provides a quick view of overall sales performance through KPIs.

Business Value

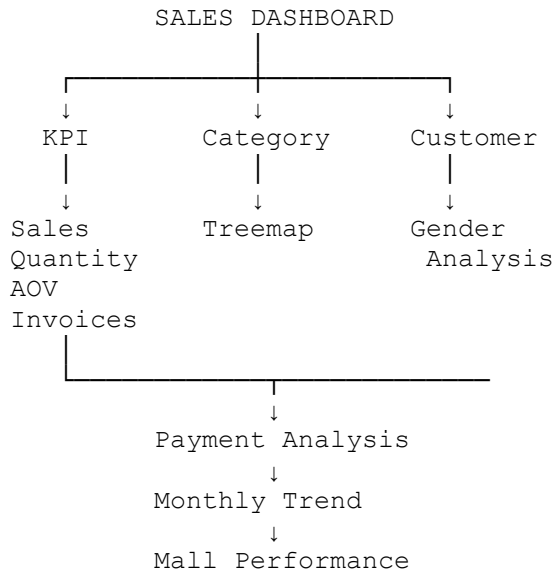
This dashboard can help a retail business answer questions such as:

- What are our total sales?
- What is the average order value?
- Which category generates the highest sales?
- Which shopping mall performs best?
- Which gender contributes more to sales?
- Which payment method is most commonly used?
- How are sales changing over time?
- Which locations require more attention?
- Where should management focus sales and marketing efforts?

Dashboard Flow

```
Raw Customer Shopping Data
↓
Data Cleaning & Preparation
↓
Tableau Data Connection
↓
Dimensions & Measures
↓
Calculated Fields
↓
KPI Creation
↓
Individual Visualizations
↓
Dashboard Design
↓
Filters & Interactions
↓
Business Insights
```

Dashboard Components



Skills Demonstrated

- Tableau
- Data Visualization
- Business Intelligence
- Dashboard Development
- KPI Development
- Sales Analysis
- Customer Analysis
- Trend Analysis
- Data Storytelling
- Interactive Reporting
- Business Problem Solving

Project Outcome

The project successfully transforms raw customer shopping data into an **interactive Tableau sales dashboard**. It provides management with a consolidated view of **sales performance, customer behavior, payment preferences, product categories, monthly trends, and shopping mall performance**.

Conclusion

The **Sales Analysis Dashboard Using Tableau** demonstrates how Tableau can be used to transform raw retail data into an interactive business intelligence solution. The dashboard combines **KPIs, treemaps, pie charts, line charts, and bar charts** to provide clear and actionable sales insights.

This project is suitable for showcasing skills relevant to **Data Analyst, Business Analyst, BI Analyst, and Tableau Developer** roles.

Project Structure

```
Tableau Sales Analysis/  
├── Tableau Workbook (.twb / .twbx)  
├── Dataset  
├── Dashboard Screenshot  
└── Project Documentation
```

Tools Used

Tableau Public | Data Visualization | Calculated Fields | Dashboard | KPI | Excel/CSV